

Analysis of Household Surveys

A five-session workshop

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Department of Economics, University of Ghana, hosted by the International Growth Centre. Five meetings of two hours, 11am–1pm, in CCC.

1 Administrivia

1.1 Meetings

Session	Day	Date	Topic
1	Tuesday	8 September	Design, sampling, and collection
2	Thursday	10 September	Describing welfare
3	Monday	14 September	Demand and welfare
4	Tuesday	15 September	From cross-sections to dynamics
5	Thursday	17 September	Capstone and frontiers

1.2 Texts

Angus Deaton's *The Analysis of Household Surveys: A Microeconomic Approach to Development Policy* (Johns Hopkins, 1997) is the spine of the workshop. The World Bank's 2018 reissue is open access, so it costs you nothing; the workshop page carries the link. Chapter 1 repays reading before the first session.

Two supplements, each for a particular session:

- Deaton & Zaidi, *Guidelines for Constructing Consumption Aggregates for Welfare Analysis* (World Bank LSMS Working Paper 135, 2002). Session 2.
- Deaton (1985), "Panel data from time series of cross-sections," *Journal of Econometrics* 30:109–126. Session 4.

1.3 Software

The hands-on work runs on a shared JupyterHub at hhsurveys.ligonresearch.org. You will be given a username and set your own password at first login. Nothing needs to be installed to take part.

The computation uses `LSMS_Library`, which gives one interface to the household surveys of some forty countries. Installing it on your own machine is worth doing — it is what you will use after the workshop ends — and the workshop page has instructions that assume no prior Python. Do it at home: it is a 1.3 GB download, and twenty laptops fetching it at once over classroom wifi will simply stall.

1.4 What you should be able to do afterwards

Not "know about" — do. By the end of the fifth session you should be able to take an unfamiliar LSMS-style survey and:

1. read its documentation well enough to say what population it represents, how it was drawn, and which of its variables the design lets you use;
2. construct a defensible consumption aggregate from its modules, deflate it across space and time, and put an equivalence scale on it;
3. produce weighted, correctly-clustered estimates of poverty and inequality, with standard errors you are willing to defend;
4. estimate Engel curves and a simple demand system, and say what they imply for the welfare consequences of a price change;
5. build a pseudo-panel from repeated cross-sections, and know when to prefer it to a real panel and when not;
6. hand the whole thing to a colleague as code that runs.

1.5 Assessment

There is none: this is a workshop, not a course. The fifth session is given over to a small analysis of a survey of your own choosing, which you will present to the room in about five minutes. Come to session 4 with a candidate country in mind.

2 Sessions

2.1 Design, Sampling, and Collection

What a careful analyst should know before touching the data.

2.1.1 Reading

- Deaton ch. 1 — the whole chapter
- Deaton §1.4 on survey design and §2.2 on what to do about it
- Skim the GLSS7 *Report* methodology section

2.2 Describing Welfare

Constructing a consumption aggregate, and then describing the distribution of it, before modelling anything.

2.2.1 Reading

- Deaton, chs. 3 and 4
- Deaton & Zaidi, *Guidelines for Constructing Consumption Aggregates* — the operational reference, and the one you will actually reach for
- Deaton & Zaidi §§2–4 if you read nothing else

2.3 Demand and Welfare

Where household surveys earn their keep for policy: what happens to whom when a price moves.

2.3.1 Reading

- Deaton, chs. 4 (§4.2 on Engel curves) and 5 (all)
- Deaton & Muellbauer (1980), "An Almost Ideal Demand System," *AER* 70:312–326
- Lewbel (1991), "The rank of demand systems," *Econometrica* 59:711–730
- Ligon, "Household Welfare from Disaggregate Demands" (working paper) — the CFE system we estimate today

2.4 From Cross-Sections to Dynamics

Two ways to get change out of survey data: build a panel from cohorts, or use one that already exists.

2.4.1 Reading

- Deaton (1985), "Panel data from time series of cross-sections"
- Deaton, ch. 2 §2.7 and ch. 6
- Abadie, Athey, Imbens & Wooldridge (2023), "When should you adjust standard errors for clustering?" *QJE* 138:1–35

2.5 Capstone and Frontiers

Where the methods break down, and then your own analysis.

2.5.1 Reading